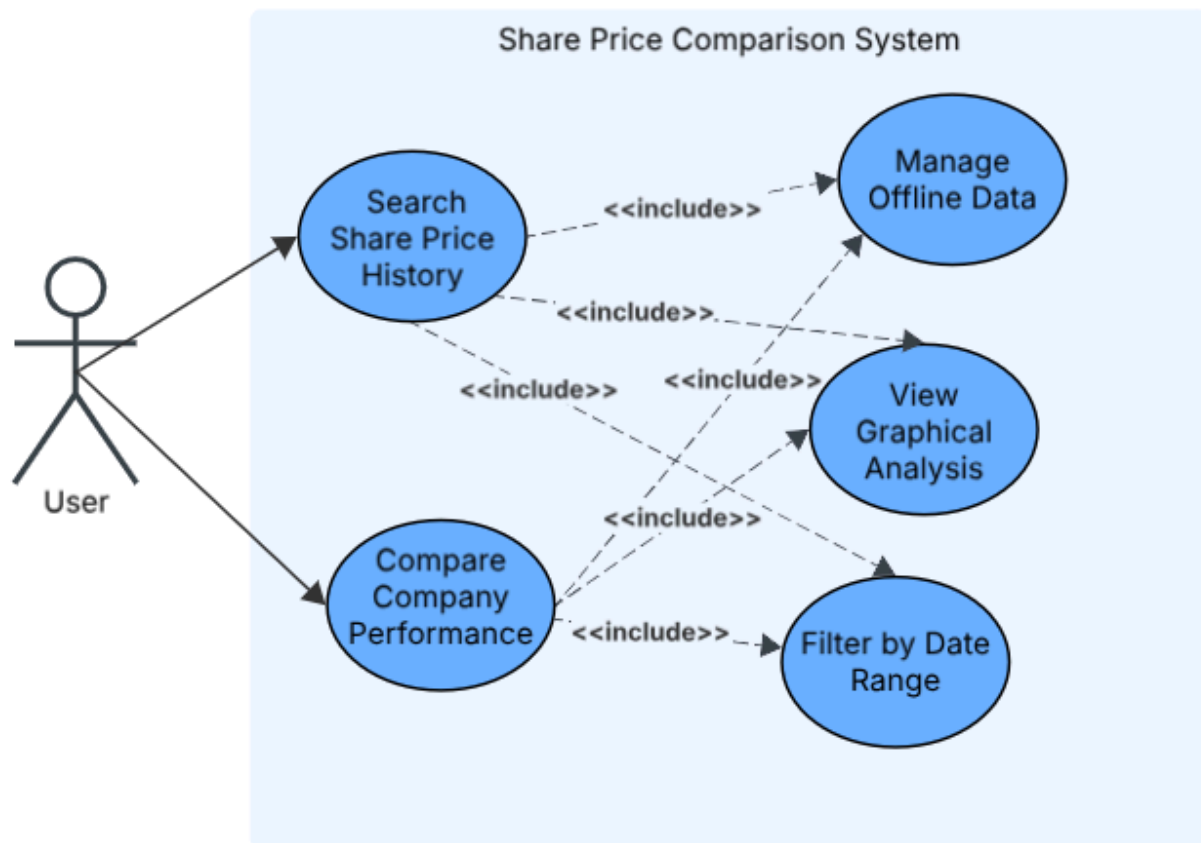


Use Case Model



Primary Actor

The User interacts with the system to analyse financial data.

Core interactions

The 'Search Share Price History', allows the user to obtain daily price information for a specific symbol.

For the 'Compare Company Performance', it enables the comparison of share performance between two different companies.

Included Behaviours(<<include>>)

The Use Case, 'Filter by Date Range', both search and comparison require a date range input, which must be limited to a maximum of two years.

For the 'View Graphical Analysis', the system must display graphs of daily share prices over time for the selected companies.

The, 'Manage Offline Data' Use Case, ensures functionality without a network connection.

The system must persistently store price data(using JSON or a relational database) whenever data is retrieved.

Use Case 1, Search Share Price History

Step	Actor Action	System Response
1	Enters a stock ticker symbol (e.g., "AAPL") and selects a start and end date.	Validates the ticker symbol format and checks that the date range does not exceed two years.
2	Confirms the search.	Checks the Local Cache (IPriceRepository) to see if data for that symbol and range already exists.
3		If the data is missing, the system calls the External API (IExternalPriceProvider) to fetch the daily prices.
4		Automatically saves any newly fetched data to the local persistence store (JSON/Database) for offline use.
5		Triggers the Graphical Analysis component to process the raw data into a time-series format.
6		Displays the share price history chart and a summary table to the Actor.

Use Case 2: Compare Company Performance

Step	Actor Action	System Response
1	Enters two different stock ticker symbols (e.g., "TSLA" and "F").	Validates both symbols and ensures the selected date range is within the two-year limit.
2	Selects the "Compare" option.	Initiates the data retrieval process for both symbols simultaneously.
3		For each symbol: Checks local storage first, then fetches from the remote provider if necessary (Manage Offline Data).
4		Normalizes the data (ensures both datasets cover the exact same dates for a fair comparison).
5		Generates a Comparison Chart showing both companies' price fluctuations on the same axes.
6		Displays the comparative graph and performance metrics (e.g., % change) to the Actor.